CSCI 113 - LAB 9; NOAH WILEY

1) word address references is 1, 4, 6, 5, 20, 17, 18, 56, 9, 11, 4, 43, 5, 6, 9, 17, 22, 27

Address	Binary Address	Cache H/M	Assigned Cache Block				
1	1	Miss	1				
4	100	Miss	1	0	0		
6	110	Miss	1	1	0		
5	101	Miss	1	0	1		
20	10100	Miss	1	0	1	0	0
17	10011	Miss	1				
18	10010	Miss	1	0			
56	111000	Miss	1	0	0	0	
9	1001	Miss	1	0	0	1	
11	1011	Miss	1	0	1	1	
4	100	Miss	1	0	0		
43	101011	Miss	1	0	1	1	
5	101	Hit	1	0	1		
6	110	Hit	1	1	0		
9	1001	Hit	1	0	0	1	
17	10001	Hit	1				
22	10110	Miss	1	1	0		
27	11011	Miss	0	1	1		

Total # of hits is 4 ⇒ 5, 6, 9, 17

Final Contents

index	contents
0	∅
1	mem [7]
2	mem [18]
3	∅
4	mem [4]
5	mem [5]
6	mem [22]
7	∅
8	mem [56]
9	mem [9]
10	∅
11	mem [27]
12	∅
13	∅
14	∅
15	∅

2) For a cache with 16 words and 2 blocks per set

16/2 = 8 sets

Set = address mod 8

Given Addresses: 1, 4, 6, 5, 20, 17, 18, 56, 9, 11, 4, 43, 5, 6, 9, 17, 22, 27

Cache Size: 16 words

Associativity: 2 blocks per set

# of Sets: 8

Replacement Policy: LRU

All sets are initally empty

Hits: 4 hits (addresses: 5, 6, 9, 17)

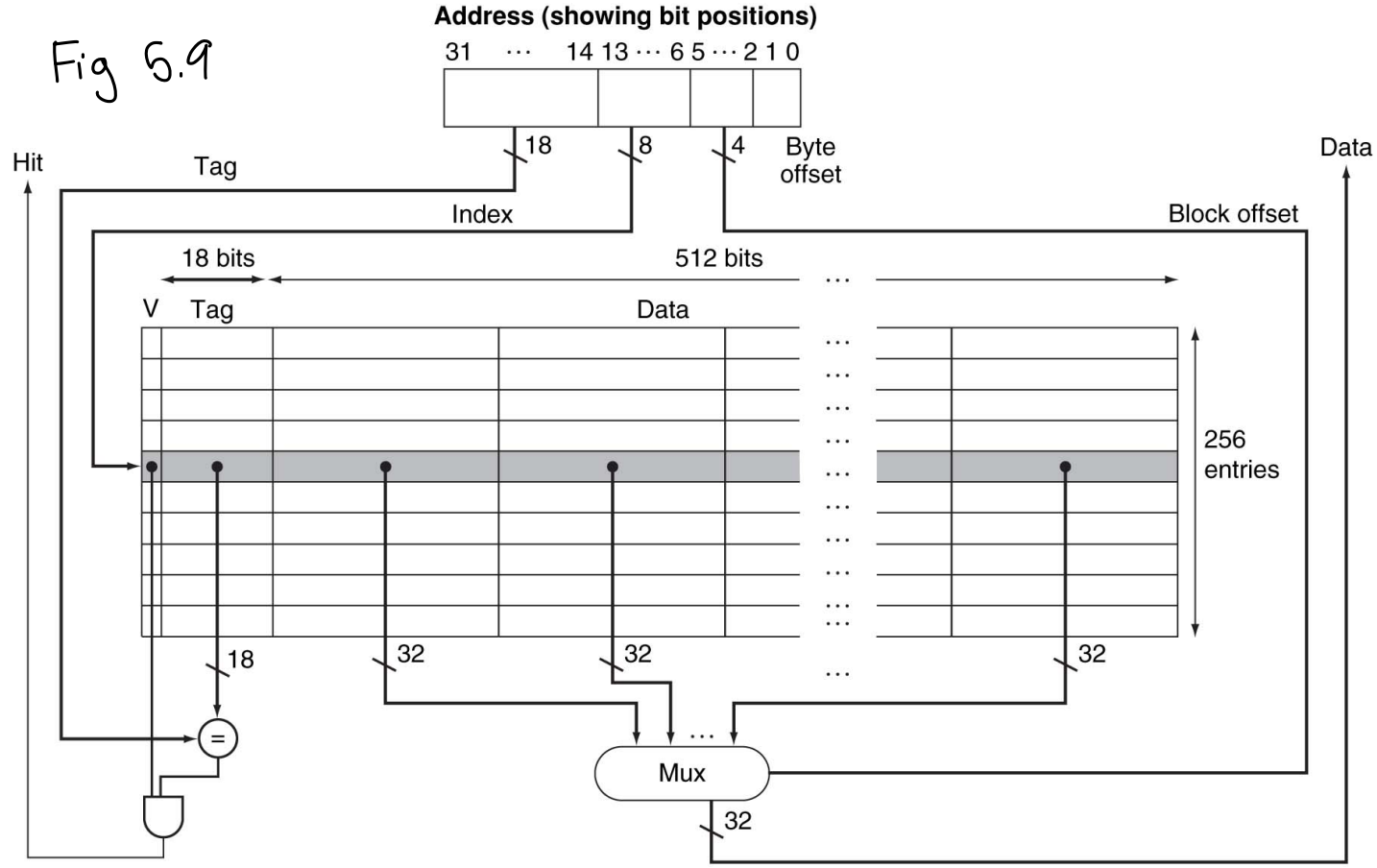
Final Cache Contents

Set	Block 0	Block 1
Set 0	mem[56]	empty
Set 1	mem[9]	mem[17]
Set 2	mem[18]	empty
Set 3	mem[27]	mem[43]
Set 4	mem[4]	mem[20]
Set 5	mem[5]	empty
Set 6	mem[22]	empty
Set 7	empty	empty

Address	Set (mod 8)	Hit / miss	Action
1	1	Miss	Place mem[1] in Set1 block 0
4	4	Miss	Place mem[4] in Set4_block 0
6	6	Miss	Place mem[6] in Set6_block 0
5	5	Miss	Place mem[5] in set 5_block 0
20	4	Miss	Place mem[20] in set 4_block 1 (LRU: mem[4])
17	1	Miss	Place mem[17] in set 1_block 1 (LRU: mem[1])
18	2	Miss	Place mem[18] in set 2_block 0
56	0	Miss	Place mem[56] in Set0_block 0
9	1	Miss	Replace mem[1] with mem[9] in Set3_block 0
11	3	Miss	Place mem[11] in Set3_block 0
4	4	Miss	Replace mem[20] with mem[4] in set4_block1
43	3	Miss	Place mem[43] in Set3_block 1
5	5	Hit	No Change
6	6	Hit	No Change
9	1	Hit	No Change
17	1	Hit	No Change

22	6	Miss	Replace mem[6] with mem[22] in set 6_block 0
27	3	Miss	Replace mem[11] with mem[27] in set 3_block 0

3) Fig 5.9



a.

Tag: 18 bits

Index: 8 bits (256 entries)

Block Offset: 4 bits (16 bytes per block; $2^4=16$)

Byte Offset: 2 bits (4 bytes per word; $2^2=4$)

Total address size = 32 bits

Number of blocks = 256

Data per block: 16 bytes = 16×8 = 128 data bits per block

Tag per Block: 18 bits

Valid Bit per Block: 1 bit

Data Storge: 256 · 128 bits = 32,768

⇒ Tag Storge: 256 · 18 bits = 4,608

Valid Bit Storage: 256 · 1 bit = 256

Total = Data + Tag + Valid

32,768 + 4,608 + 256 = 37,632 bits

b. 4 words/block · 4 bytes/word = 16 bytes/block

Byte offset: each word is 4 bytes ($2^2=4$), the byte offset remains 2 bits

Block Offset: a block now has 16 bytes ($2^4=16$), so the block offset is 4 bits

Index: Orginal Configuration: 256 blocks ($2^8=256$)

New number of blocks $= \frac{256}{4} =$ 64 blocks

index field is now $\log_2(64) =$ 6 bits

Tag: Tag bits $= 32-(6+4+2) =$ 20 bits

Total Cache Storage $= (128+20+1) \times 64 = 149 \times 64 =$ 9536 bits

Tag: 20 bits
Index: 6 bits
Block Offset: 4 bits
Byte offset: 2 bits
Total cache storage: 9,536 bits

4) Formula: Miss Penalty = Memory Latency + (# of Words per Block · Transfer Time per Word).

$$\text{Memory Bandwidth} = \frac{\text{Block Size (in words)}}{\text{Total Time Taken to Transfer the Block}}$$

Organization: One-Word-Wide Memory

Miss Penalty:

- Memory Latency = 20 cycles
- Block Size = 20 words

Miss Penalty = $20 + (20 \cdot 2) = 20 + 40 = 60$ Cycles

Memory Bandwidth:

- Miss Penalty = 60 Cycles
- Block Size = 20 words

$$\text{Memory Bandwidth} = \frac{20 \text{ words}}{60 \text{ cycles}} = \frac{1}{3} \text{ words / cycles}$$

Organization: Wider Memory (4 words)

Miss Penalty:

- Memory Latency = 20 cycles
- Block Size = 20 words
- Each transfer 4 words and takes $2 \cdot 4 = 8$ cycles per 4 words
- # of transfers = $20/4 = 5$

Miss Penalty = $20 + (5 \cdot 8) = 20 + 40 = 60$ Cycles

Memory Bandwidth:

- Miss Penalty = 60 cycles
- Block Size = 20 words

$$\text{Memory Bandwidth} = \frac{20 \text{ words}}{60 \text{ Cycles}} = \frac{1}{3} \text{ words / cycle}$$

Organization: Interleaved Memory (4 banks)

Miss Penalty:

- Memory Latency = 20 Cycles
- Block Size = 20 words
- Transfer time for 4 words = 2 cycles
- # of transfers = 20/4 = 5

Miss Penalty = 20+(5×2) = 20 + 10 = 30 cycles

Memory Bandwidth:

- Miss Penalty = 30 Cycles
- Block Size = 20 words

Memory Bandwidth = $\frac{20 \text{ words}}{30 \text{ cycles}} = \frac{2}{3}$ words/cycles

5)

Given: Virtual Address: 42 bits (byte addressable)

Page Size: 8KB (2^{13} bytes per page)

Physical Address: 32 bits (byte addressable)

Page Table Entry (PTE): 4 bits

Page Size = 8KB = 2^{13} Bytes	Virtual Address = 42 bits	Physical Address = 32 bits
Page Offset = 13 bits	VPN = 42 − 13 = 29 bits	PFN = 32 − 13 = 19 bits

PFN: 19 bits

Other fields: 4 bits

$\Rightarrow$ PTE = 19 + 4 = 23 bits

# of virtual pages = 2^{VPN}

VPN = 29 bits, so: 2^{29}

Each Entry = 23 bits

Total Size = 23 bits $\times$ 2^{29} entries

Total size (bytes) = $\frac{23 \cdot 2^{29}}{8} \Rightarrow = 2.875 \times 2^{29}$ bytes

2^{29} bytes = 2^{29-30} GB = 0.5 GB

Total size = 2.875 $\times$ 0.5 GB = 1.4375 GB